GAME THEORY:

FUNDAMENTALS

Session 12

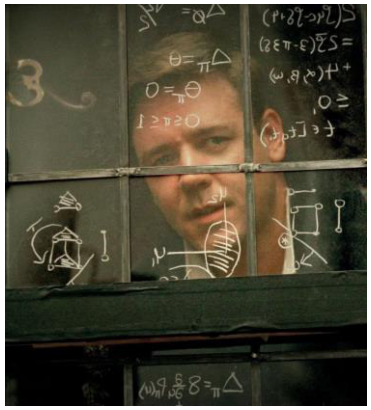
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Agenda for today: Game theory

Game theory **are (Mathematical) models to study the strategic interaction of rational decision-makers**

Your rivals' optimal choices depend on their expectations of what you intend to do, which, in turn, depends on their assessments of your assessment about them.



Agenda for today

1. Introduction to Game Theory?

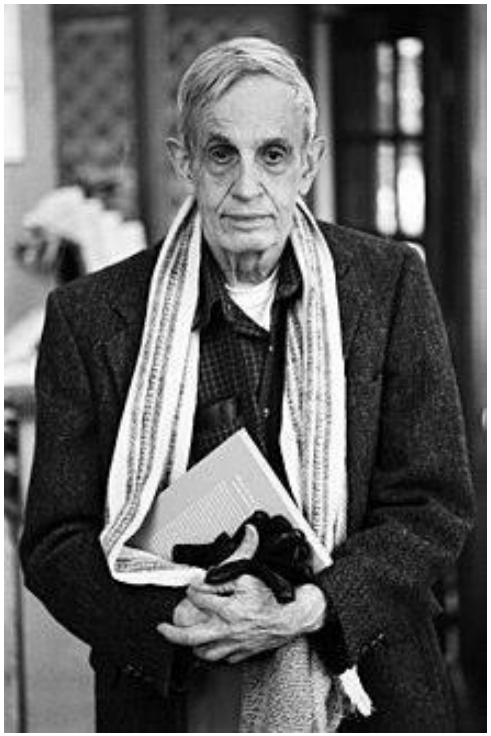
2. Dominant Strategy Equilibrium

3. Dominated Strategy

4. Nash Equilibrium

5. Subgame Perfect Nash Equilibrium

John Nash



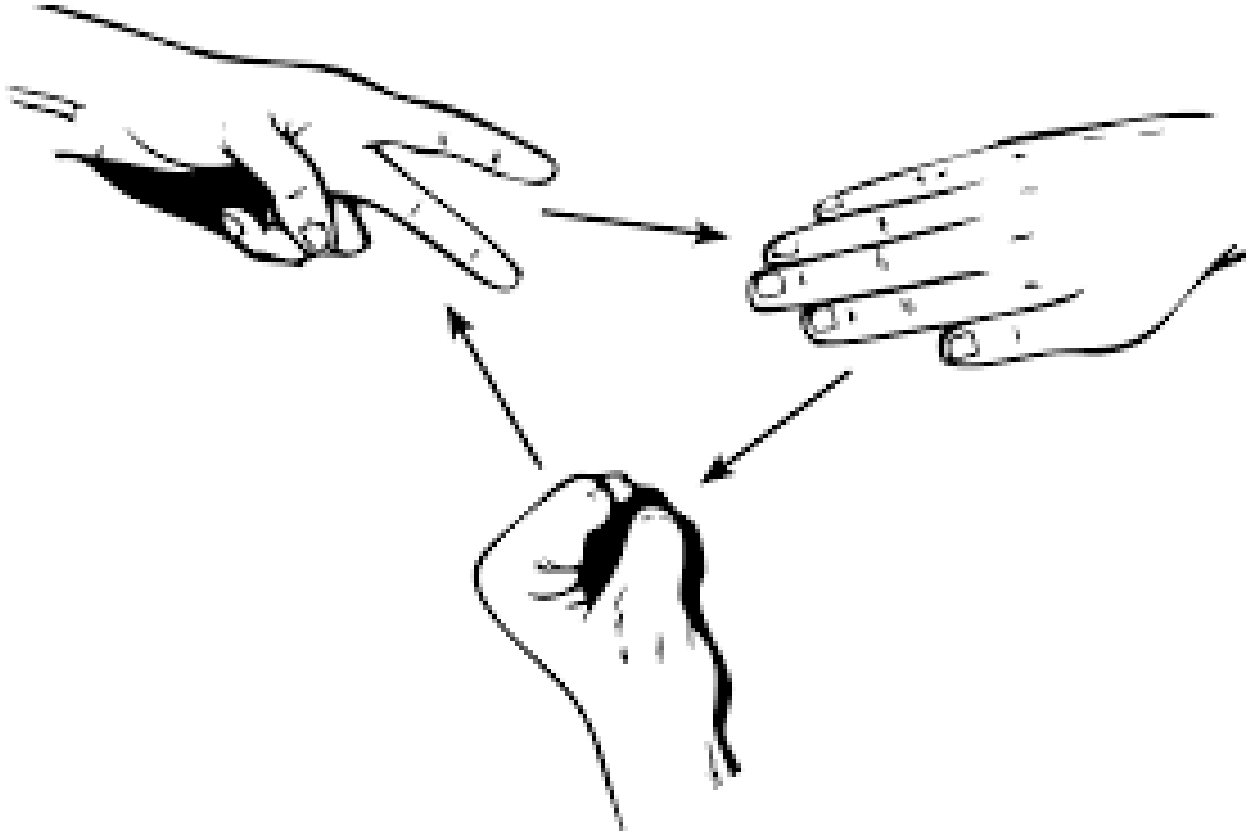
Nobel prize 1994 for his work on game theory in non-cooperative games

→ Nash equilibrium

Game Theory and Nobel Prize Winners

- 1994 : John Nash, Reinhard Selten, John Harsanyi
- 1996 : William Wickrey and James Mirrlees
- 2002: D. Kahneman
- 2005 : Robert Aumann, Thomas Shelling
- 2007 : Leonid Hurwicz, Erick Maskin, Roger Myerson
- 2011: Spence, Stiglitz, Akerlof
- 2012 : Lloyd Shapley, Alvin Roth
- 2014: Jean Tirole
- 2015: Deaton
- 2016: Holmstrom & Hart
- 2017: Thaler

Scissors – Paper – Stone

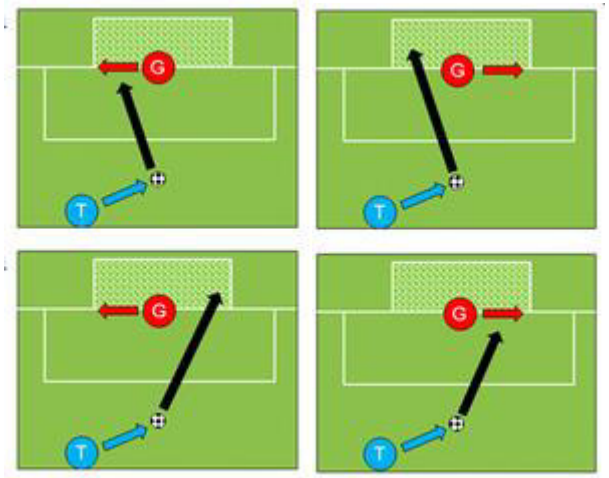


The game



A set of payoffs to each player for each possible configuration of strategies

A set of Players



A set of available strategies for each player



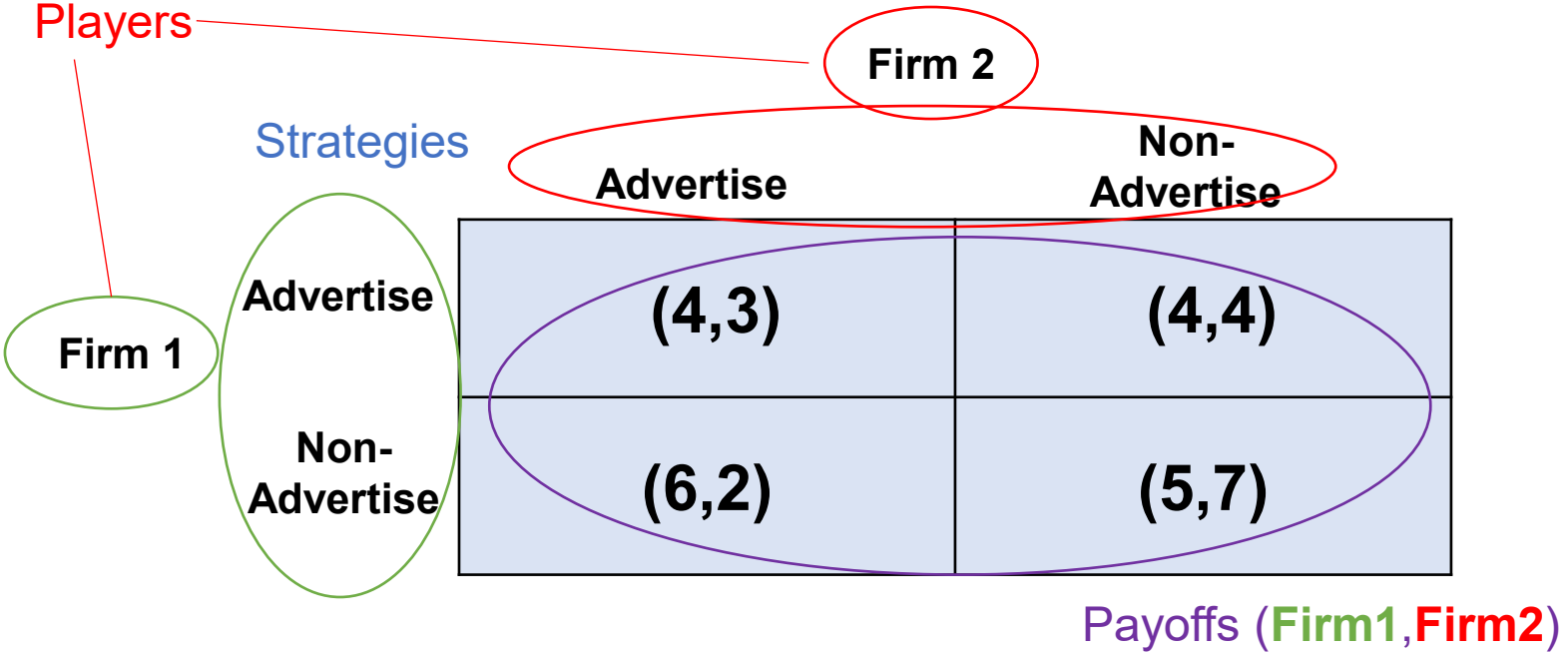


Why is game theory useful for strategic choices?

1. Incorporate the reactions of rivals into your strategic options
2. Assign probabilities to the actions / reaction
3. Trying to anticipate moves
4. Find the optimal or best choice (first best vs second best)

How can a game be illustrated?

Payoff Matrix – Normal Form



Important: **Game theory assumes that players always act rationally (they do what is best for them, but also they assume that their competitors are rational)**

There are four main concepts to solve strategic games:

1) **Dominant Strategies** & Dominant Strategy Equilibrium
2) **Dominated Strategies** & Iterative Elimination of Dominated Strategies
3) **Nash Equilibrium**
4) **Backward Induction** and **Subgame Perfect NE**

1. Introduction to Game Theory?

2. Dominant Strategy Equilibrium

3. Dominated Strategy

4. Nash Equilibrium

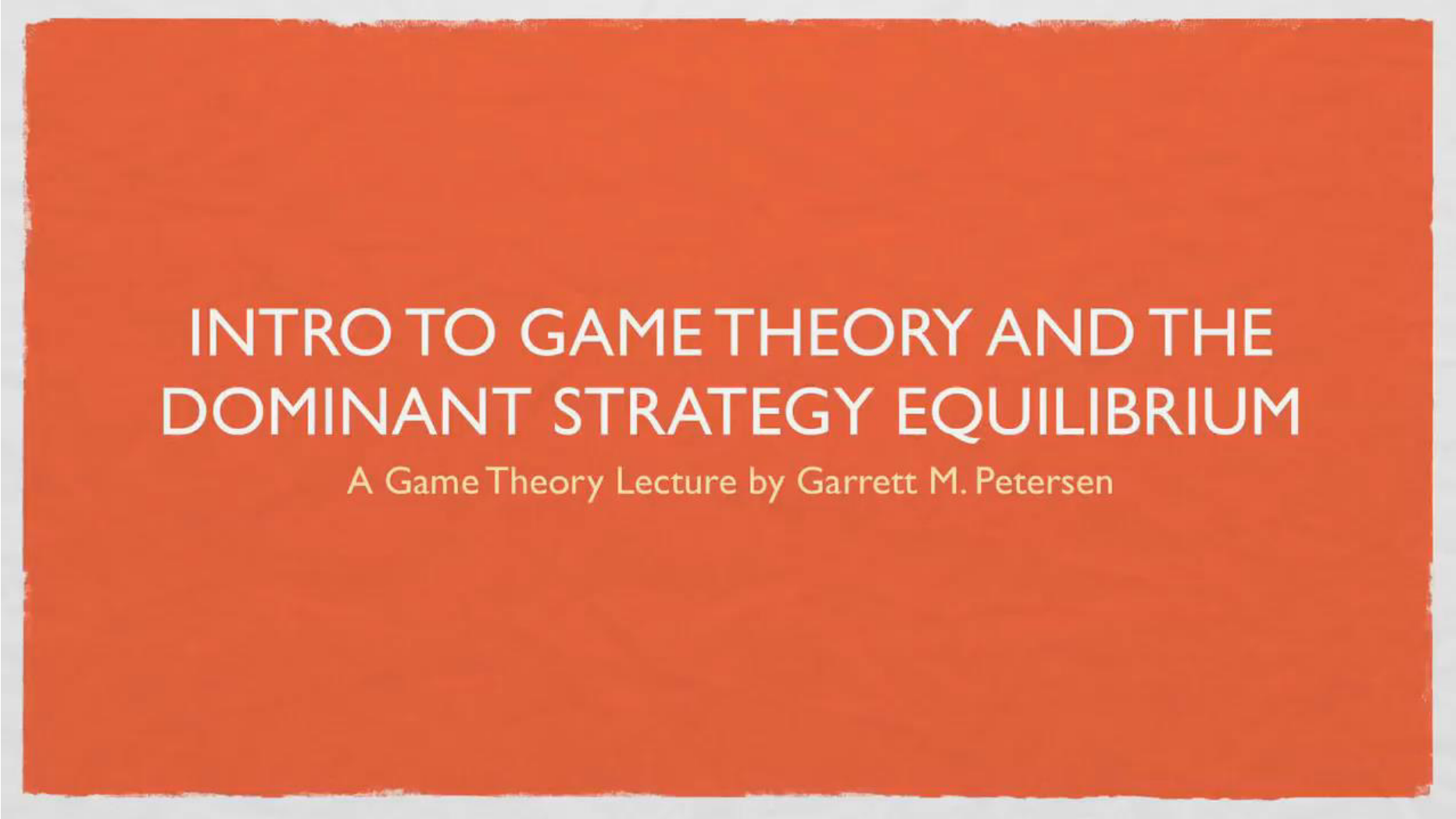
5. Subgame Perfect Nash Equilibrium

1. Dominant strategy equilibrium

- A strategy is a **Dominant strategy** for a player if it yields the best payoff (for that player) regardless what strategies the other players choose
  - *Whenever a player has a strategy that is strictly better than any other strategy regardless of the other players' strategy choices, we say that the first player has a **dominant strategy**.*

- **If all players have a dominant strategy, they will all choose their dominant strategy. This leads to a dominant strategy equilibrium.**

Intro to Game Theory and the Dominant Strategy Equilibrium (4m)
https://www.youtube.com/watch?v=3Y1WpytiHKE



1. Dominant strategy

Remember:

(**Payoff Firm 1**, **Payoff Firm 2**)

For **Firm 2** it is always better to **not advertise**
(no matter what Firm 1 does)
Dominant Strategy → Non Advertise

6>4
5>4

4>3
7>2

Firm 1 \ Firm 2	Advertise	Non-Advertise
Advertise	(4,3)	(4,4)
Non-Advertise	(6,2)	(5,7)

For **Firm 1** it is always better to **not advertise**
(no matter what Firm 2 does)
Dominant Strategy → Non Advertise

Equilibrium: (Non Advertise, Non Advertise)

(5 , 7)

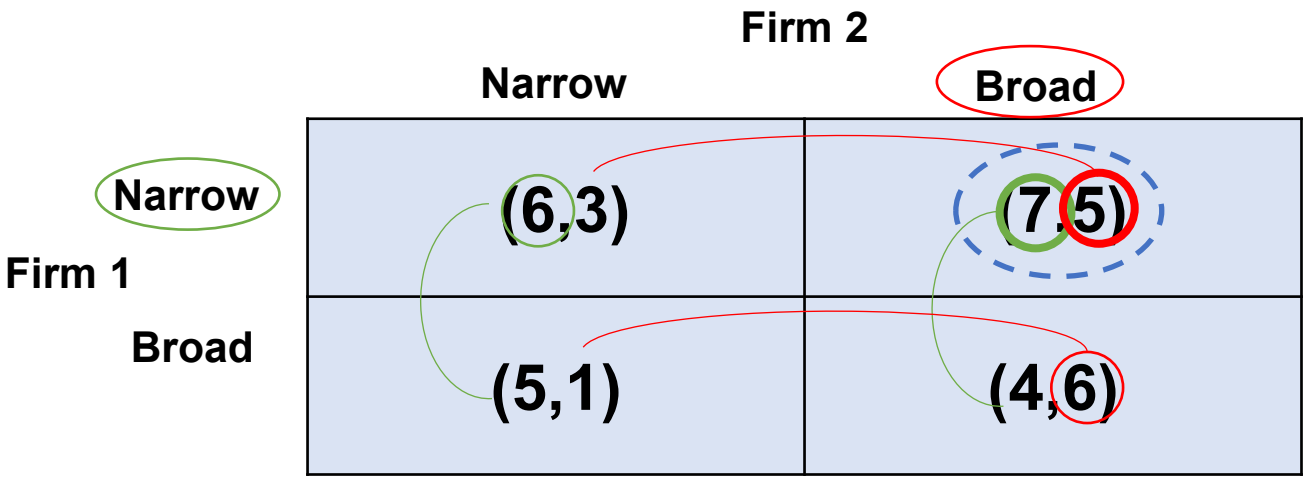
Example: What is the Dominant Strategy Equilibrium?

Two competitors have to decide about the coverage of market segments (**narrow vs. broad strategy**).

After market analysis they estimate the following pay-offs for each strategy:

6>5
7>4

5>3
6>1



Equilibrium: (narrow, broad) (7,5)

Example: What is the Dominant Strategy Equilibrium?

2 > 1 ; 0
5 > 4 ; 1
1 > 0 ; 0
4 > 2 ; 3

6 > 2 ; 5 ; 1
3 > 1 ; 0 ; 2
4 > 3 ; 3 ; 1

Broad

Firm 2

Firm 1	**Expand High**	**Expand Medium**	**Expand Small**	**Expand International**
Expand High	2,2	5,6	1,5	4,1
Expand Medium	1,2	4,3	0,0	2,2
Expand Small	0,3	1,4	0,1	3,1

Equilibrium: (Expand High, Expand Medium) (5,6)

What if there is no dominant strategy?

5 > 3 ; 2 ; 4
5 > 4 ; 3 ; 4

Player 1	Player 2: X	Player 2: Y
A	5,2	4,2
B	3,1	3,2
C	2,1	4,1
D	4,3	5,4

2 = 2
2 > 1
1 = 1
4 > 3

1. Introduction to Game Theory?

2. Dominant Strategy Equilibrium

3. Dominated Strategy

4. Nash Equilibrium

5. Subgame Perfect Nash Equilibrium

2. Dominated strategy

A strategy is dominated for a player if the player has another strategy that performs at least as good (or better) than the dominated strategy no matter what other players choose.

- In case of a dom**inant** strategy all other strategies of a player are dom**inated**.

- But there are cases where a player does not have a dom**inant** strategy, however has dom**inated** strategies.

2. Dominated strategy - Example

5 > 3 ; 2 ; 4
5 > 4 ; 3 ; 4

		Player 2 X	Player 2 Y
Player 1	A	5,2	4,2
	B	3,1	3,2
	C	2,1	4,1
	D	4,3	5,4

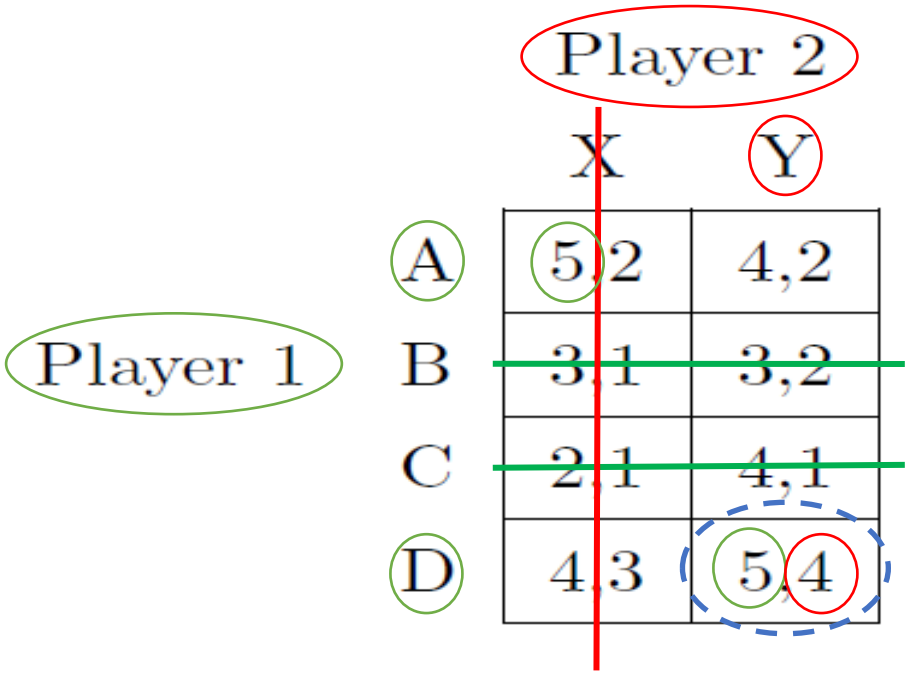
2 = 2
2 > 1
1 = 1
4 > 3

Player 1: no single dominant strategy that 'beats' all the other strategies

Player 2: **weakly** dominant strategy (Strategy Y **NEVER** equally good or better than X)

2<5 2<4
1<3 2<3
1<2 1<4
3<4 4<5

2. Dominated strategy - Example



- B & C are dominated strategies for Player 1 (A & D are always equally good or better no matter what Player 2 does)
- X is a dominated strategy for Player 2
- Player 1 assumes that Player 2 will never choose X
- Player 2 assumes that Player 1 will not choose B or C

2. Dominated strategy - Example

- This procedure is called:

  ITERATED ELIMINATION OF DOMINATED STRATEGIES

- Therefore the game reduces to

5 > 4

4 > 2

Player 1 \ Player 2	Y
A	4,2
D	5,4

In the reduced game:

- Player 1 → D dominates A
- Player 2 → Y
- Equilibrium: (D;Y)

  yielding a payoff of (5,4)

Agenda for today

1. Introduction to Game Theory?

2. Dominant Strategy Equilibrium

3. Dominated Strategy

4. Nash Equilibrium

5. Subgame Perfect Nash Equilibrium

What if we cannot solve a game by identifying dominant and dominated strategies?

There are games that cannot be solved through dominant strategies, or through iterated elimination of dominated strategies (iterated elimination does not always lead to a ***single*** strategy profile)

3. Nash equilibrium

At the Nash equilibrium both players will say:

"Given what the other player selected, I am happy with my choice".

Definition: For a two-player game, a Nash equilibrium is an outcome where player 2's strategy is the best response to player 1's strategy and player 1's strategy is a best response to player 2's strategy.

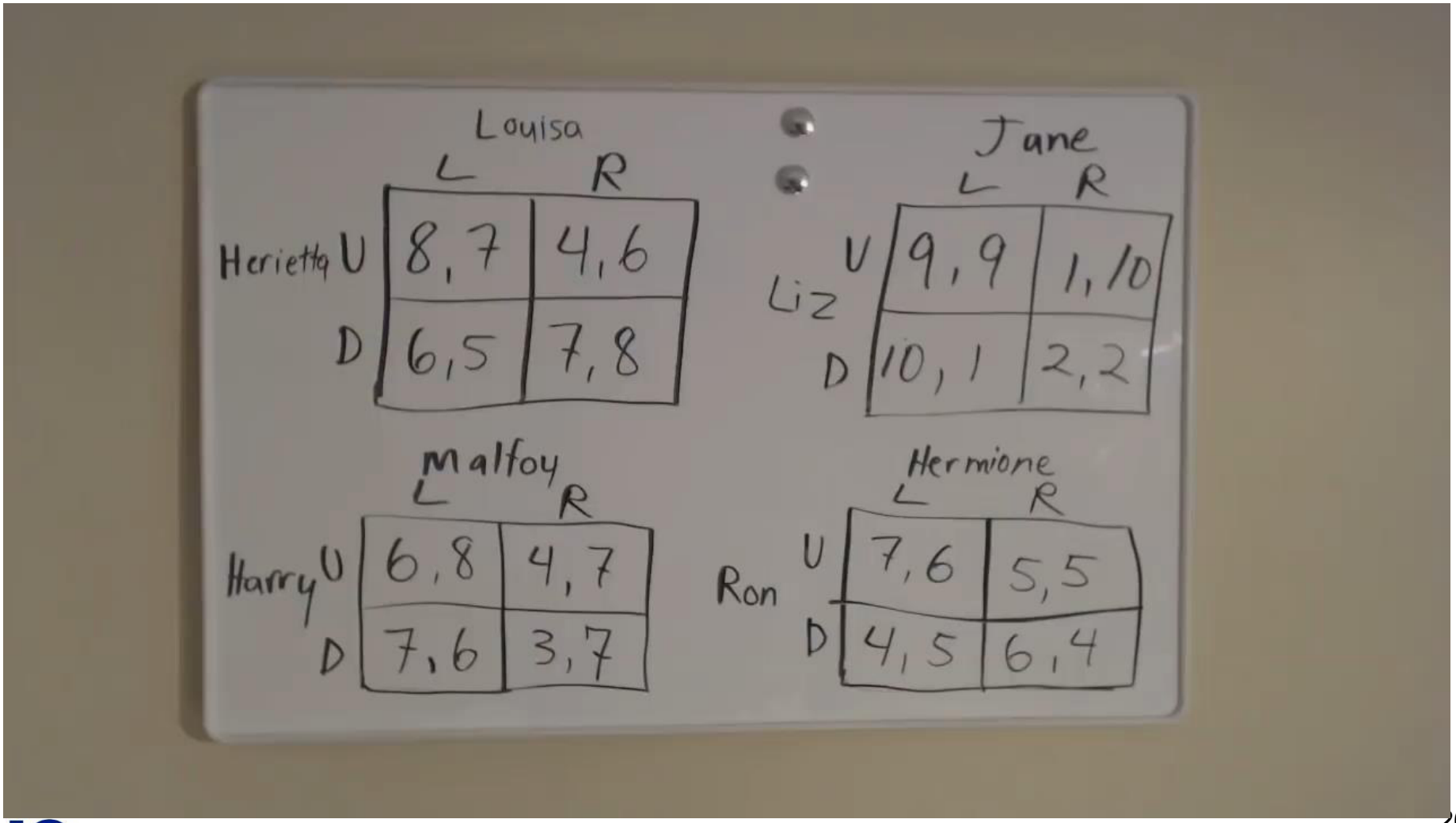
- A strategy profile is a **Nash equilibrium** if no player wants to unilaterally deviate to another strategy, given other players' strategies
- A Nash Equilibrium is a set of best-response strategies.
- In a game **can be several Nash Equilibriums**
  from 0 to n x n (all the matrix) Nash equilibriums.



Nash equilibrium Exercises

https://www.youtube.com/watch?v=6rs_EQpxTI4





Nash equilibrium Exercises

IMPORTANT → Nash Equilibrium Examples (5 mts.)
https://www.youtube.com/watch?v=6rs_EQpxTI4

Nash Equilibrium (3m) at home
https://www.youtube.com/watch?v=5o6MFTJGwuc

Exercise in class: Please find the Nash Equibrium

		Beta		
		Do Not Expand	*Small*	*Expand*
Alpha	DO NOT EXPAND	$18, $18	$15, $20	$9, $18
	SMALL	$20, $15	$16, $16	$8, $12
	LARGE	$18, $9	$12, $8	$0, $0

Exercise: Please find the Nash Equibrium

20 > 18 ; 18
16 > 15 ; 12
9 > 8 ; 0

20 > 18 ; 18
16 > 15 ; 12
9 > 8 ; 0

Nash Equilibrium

		Beta		
		Do Not Expand	*Small*	*Expand*
Alpha	DO NOT EXPAND	$18, $18	$15, $20	$9, $18
	SMALL	$20, $15	$16, $16	$8, $12
	LARGE	$18, $9	$12, $8	$0, $0

- The Nash equilibrium does not necessarily correspond to the outcome that maximizes the aggregate profit of players. **Which would be the best option for COOPERATION? (18,18) → 18+18 =36**

  Alpha and Beta would be collectively better off by cooperation an DO NOT EXPAND. However the rational pursuit of self-interest leads to take actions that is detrimental to their collective interest.
- If each party expects the other party to chose its Nash Equilibrium strategy, then both parties will chose their Nash Equilibrium strategies. At the Nash equilibrium, expectations equals outcome. Expected behavior and actual behavior converge.

Pareto optimality

Video: Pareto Optimal (4 m.)

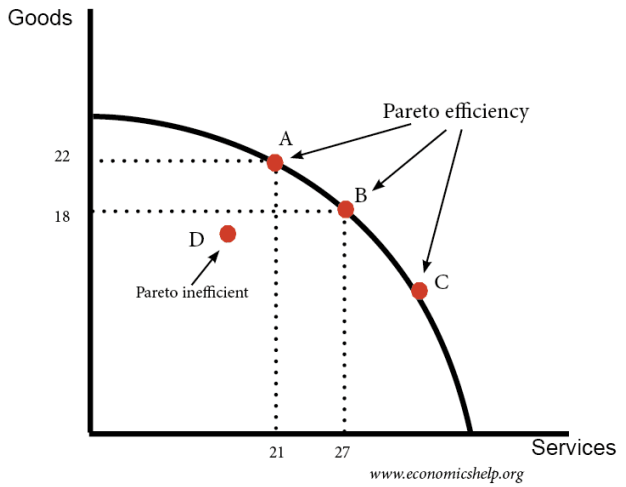
https://www.youtube.com/watch?v=idHVAUEeaqE

An outcome of a game is **Pareto optimal** if there is no other outcome that makes **every player at least as well off** and **at least one player strictly better off**.

- Pareto optimal = Pareto efficient
- **Is the Nash Equilibrium in the Prisoner's Dilemma Pareto efficient?**

NO

- An outcome is Pareto inefficient if there is another outcome that would give both players a higher payoff or that would give one player the same payoff and the other player a higher payoff



The Prisoner's Dilemma (6 mts.)

https://www.youtube.com/watch?v=t9Lo2fgxWHw

		Prisoner A Choices	
		Stay Silent	*Confess and Betray*
Prisoner B Choices	*Stay Silent*	Each serves one month in jail	Prisoner A goes free Prisoner B serves full year in jail
	Confess and Betray	Prisoner A serves full year in jail Prisoner B goes free	Each serves three months in jail

Best response for both players

0 > -30
-90 > -365

0 > -30
-90 > -365

The Prisoners' Dilemma

Nash Equilibrium

Prisoner A Choices		Prisoner B Choices: Stay silent	Prisoner B Choices: Betray
	Stay silent	(-30, -30)	(-365, 0)
	Betray	(0, -365)	(-90, -90)

Example: Price war



Example of a "prisoners' dilemma matrix applied to a economic game)

Nash Equilibrium

Best cooperation answer

5 > 0
20 > 10

5 > 0
20 > 10

pepsi

Coca-Cola

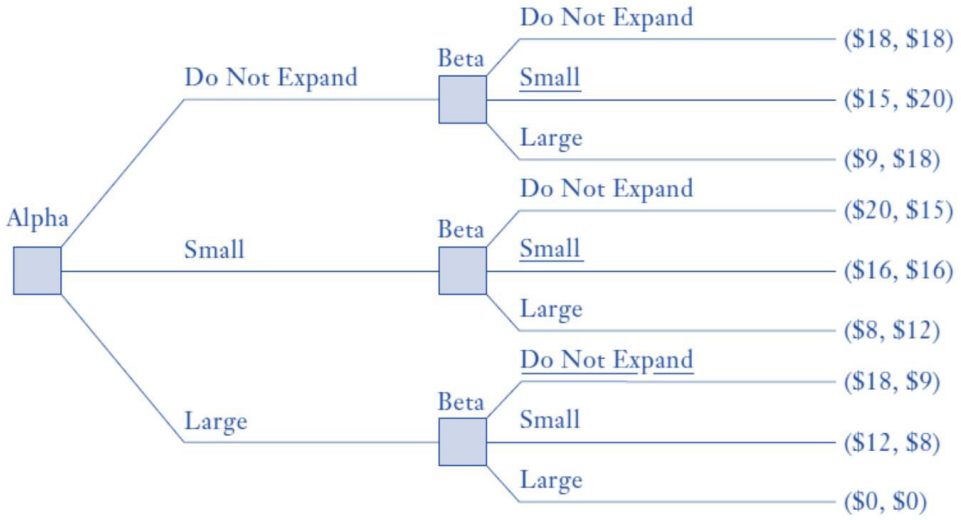
	Lower Price	Not Lower Price
Lower Price	5, 5	20, 0
Not Lower Price	0, 20	10, 10

Simultaneous vs. sequential games

- **MATRIX FORM** is useful when each party moves simultaneously

		Beta		
		Do Not Expand	*Small*	*Expand*
Alpha	DO NOT EXPAND	$18, $18	$15, $20	$9, $18
	SMALL	$20, $15	$16, $16	$8, $12
	LARGE	$18, $9	$12, $8	$0, $0

- **EXTENSIVE FORM** is useful for Sequential games and it is better represented. *(Alpha decides first; then Beta decides)*



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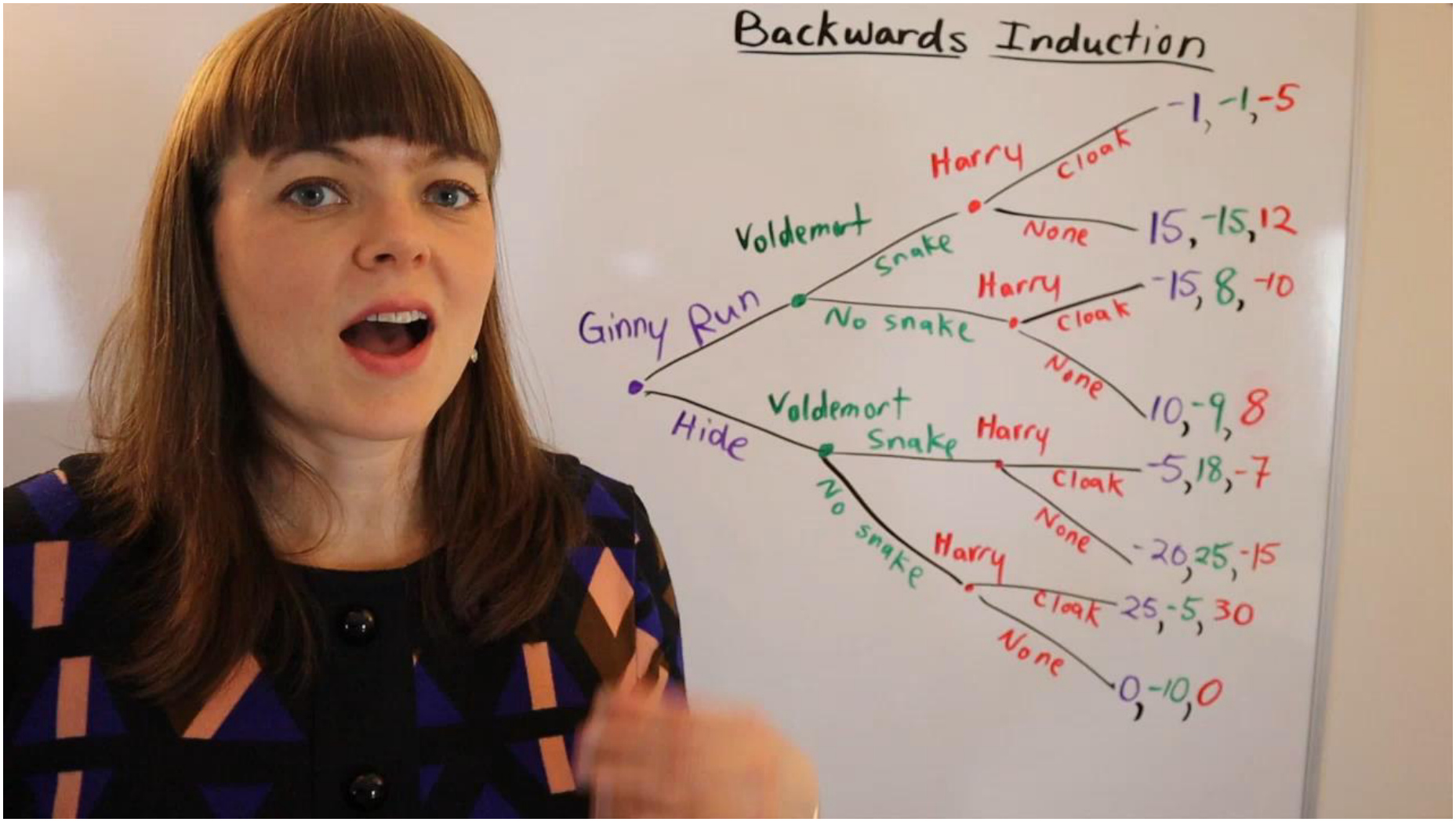
5. Subgame Perfect Nash Equilibrium

Video: Subgame Perfect Nash Equilibrium (3 m.)

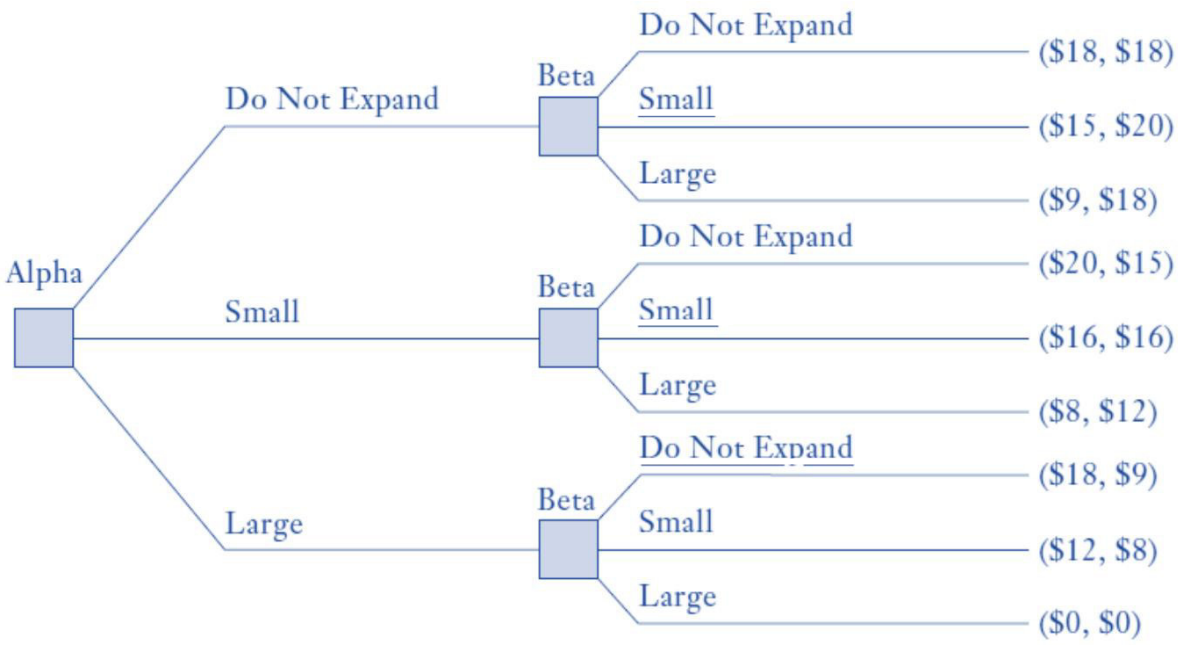
https://www.youtube.com/watch?v=dTTFzBlpdKA&list=PL_o20EJtc-0hsQ8JEDMVV9XEVVK2JAOed&index=4



Backwards Induction Game Tree (8 m.) https://www.youtube.com/watch?v=0Hk59luySfg



Extensive form



- Nb. of players: 2
- Complete information
- List of actions for each player
- List of payoffs for each player
- 2 Subgames
- Alpha decides first

Solved by backward induction

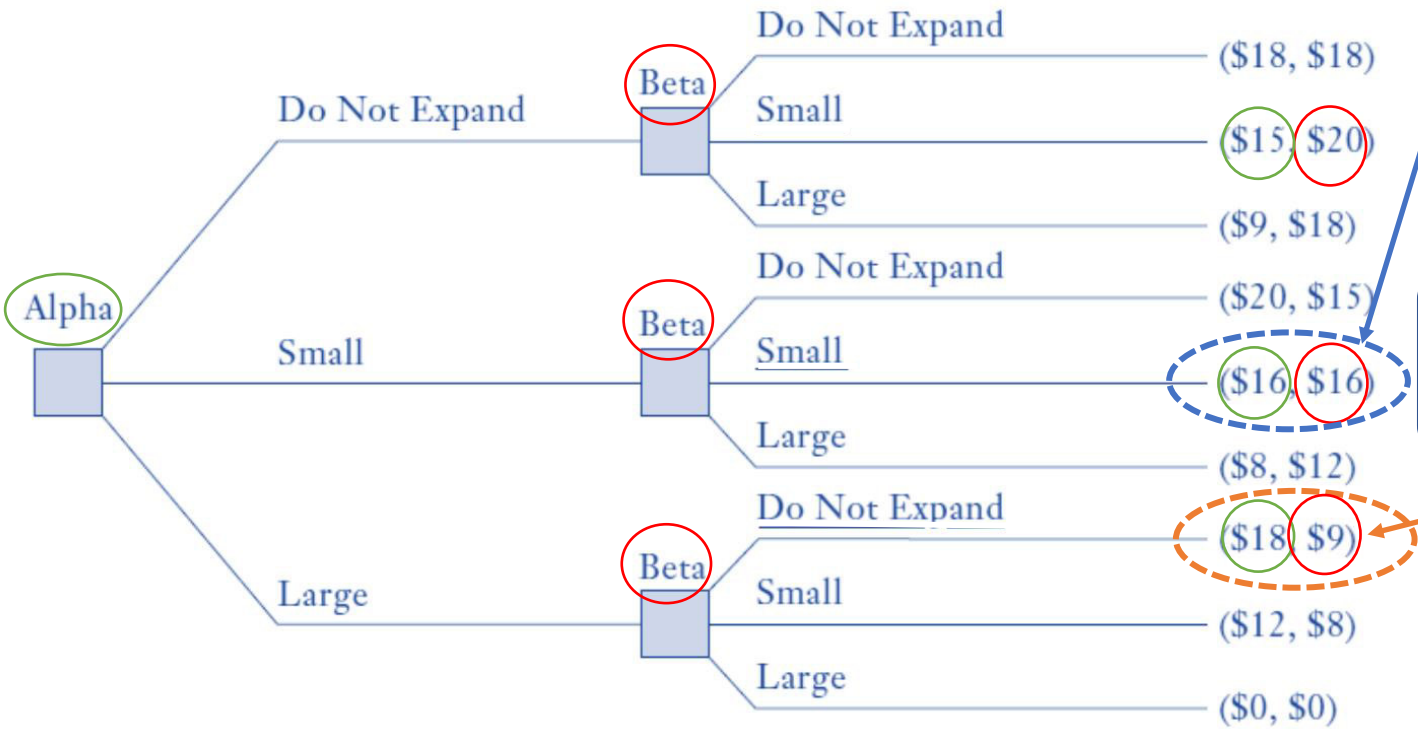
Extensive form: Backward induction

1. Start from the back
2. Find BETA's optimal decision for each of the 3 choices of ALPHA (20 , 16 & 9)
3. ALPHA have to select between the 3 best responses of Beta. Alpha will select the best for them. (18 > 16 > 15)
4. **Solution is (18,9) Subgame Perfect Nash Equilibrium**

Backward induction

20 > 18 ; 18
16 > 15 ; 12
9 > 8 ; 0

18 > 16 ; 15



Both takes the decision at the same time

Nash Equilibrium: SMALL, SMALL (16,16)

Alpha decides first and Beta adapts

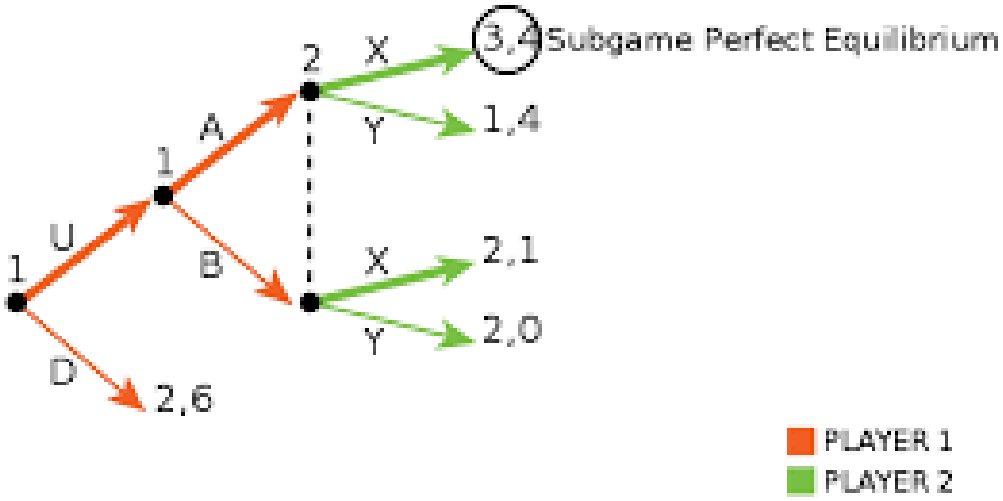
Subgame Perfect Nash Equilibrium: LARGE, DO NOT EXPAND (18,9)

Thanks a lot!

Any questions?

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Video 2 (watch at home): Subgame Perfect Nash Equilibrium (8 m.) watch at home
https://www.youtube.com/watch?v=dTTFzBlpdKA&list=RDCMUC5a61ychNIt2VI7CZ1pGnJA&index=2



1. Construct the REVENUE matrix

In a market there are two companies that sells cellular phones (Apple and Samsung). The price of each phone is $900 and the total cost is $200 for both companies.

- If Apple does HIGH investment in marketing then they will sell:
  - If Samsung does HIGH investment → Apple sell 20 K phones and Samsung 20 K.
  - If Samsung does LOW investment → Apple sell 30 K phones and Samsung 7,142 phones.
- If Apple does LOW investment in marketing then they will sell:
  - If Samsung does HIGH investment → Apple sell 7,142 phones and Samsung 30 K.
  - If Samsung does LOW investment → Apple sell 10 K phones and Samsung 10 K.

1. Construct the REVENUE matrix

$700 * 30 K = 21 M$
$700 * 20 K = 14 M$
$700 * 10 K = 7 M$
$700 * 7,142 = 5 M$

	($ million)	Samsung: High investment Marketing	Samsung: Low investment Marketing
Apple	High investment Marketing	(21 , 21)	(14 , 5)
Apple	Low investment Marketing	(5 , 14)	(7 , 7)

Which is the Nash Equilibrium?

2. SIMULTANEOSUS GAMES

Two meat analogue producers contemplate selling a new product in the same marketplace. The products sold by the two firms are similar, although firms may choose a distinctive level of quality. Each can choose to go for the high end of the market (high quality) or the low end (low quality). A marketing analyst predicts that, for all possible combinations of choices, the resulting profits in year 2021 will be as given by the following payoff matrix:

		Firm 2	
		Low quality	*High quality*
Firm 1	*Low quality*	-2, -3	80, 30
	High quality	10, 60	6, 6

2. SIMULTANEOSUS GAMES

a) What choices of quality, by individual firms, are Nash equilibria?

There are two Nash equilibria: [high quality, low quality] = (10, 60) and [low quality, high quality] = (80, 30)
In words, in the first Nash equilibrium, firm 1 finds it profitable to choose the high quality level when firm 2 chooses the low level, and simultaneously firm 1 finds it profitable to choose the low quality level when firm 2 chooses the high level.

		Firm 2	
		Low quality	*High quality*
Firm 1	*Low quality*	-2, -3	80, 30
	High quality	10, 60	6, 6

2. SIMULTANEOSUS GAMES

b) Firms may choose to cooperate by coordinating their choices, in order to maximize joint profits (that is, the sum of individual profits). In that case, what is the cooperative outcome?

Joint profits are obtained by summing firms' individual profits in each combination of possible choices. The maximum is 110=80+30, as obtained when Firm 1 offers low quality and Firm 2 high quality. This is the cooperative outcome (, which is also a Nash equilibrium in this case. But attention: This is not always the case).

2. SIMULTANEOSUS GAMES

c) Which firm benefits most from the cooperative outcome? How much would that firm need to offer the other to persuade it to coordinate choices?

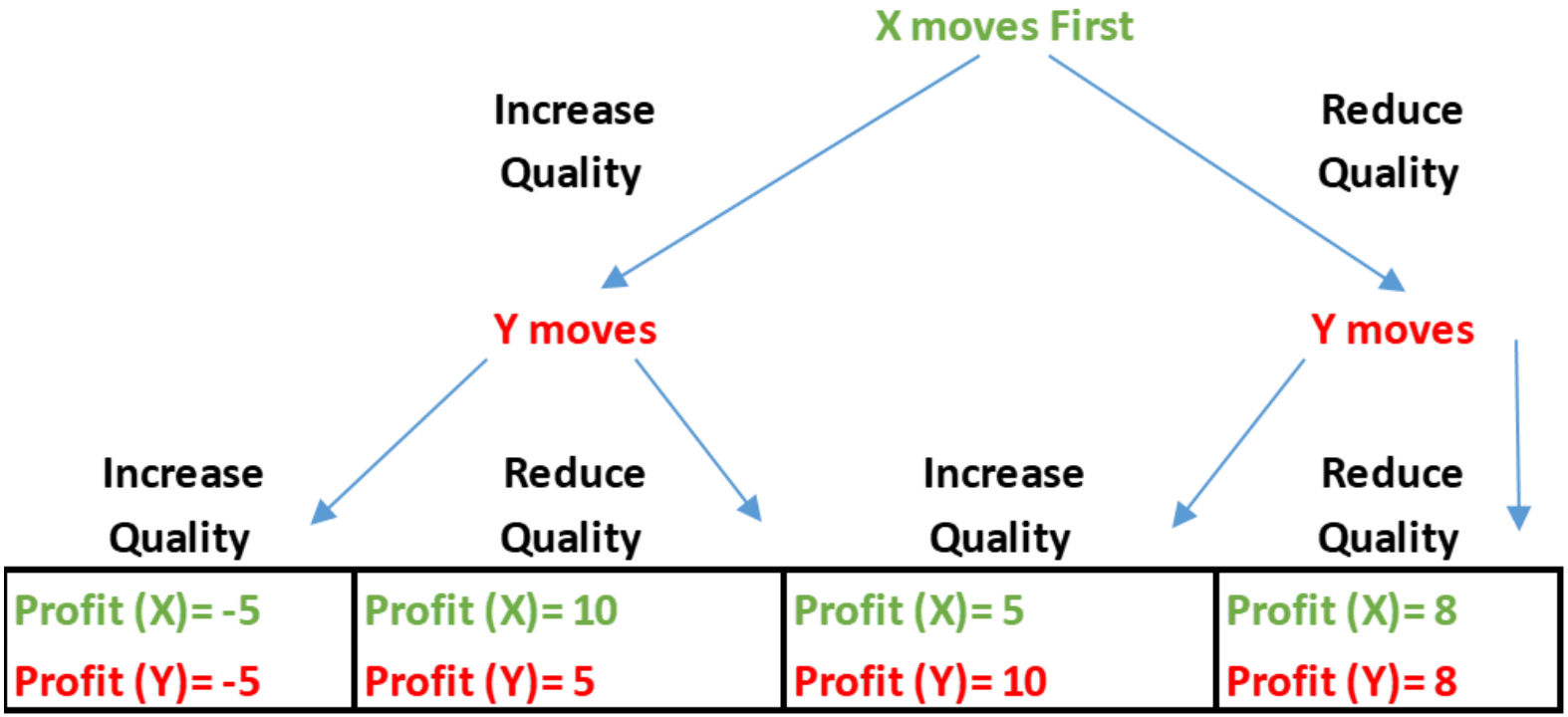
Compared to the next best opportunity, Firm 1 benefits most from the cooperative outcome [high quality, low quality], as it gains 70=80-10, whereas Firm 2 gains 30=60-30. Therefore Firm 2 will agree to be the one that chooses a high level of quality if Firm 1 "subsidizes" it by transferring 30+ε, where ε can be arbitrarily small. It remains to check whether such a cooperative agreement is legal or not.

		Firm 2	
		Low quality	*High quality*
Firm 1	*Low quality*	-2, -3	80, 30
	High quality	10, 60	6, 6

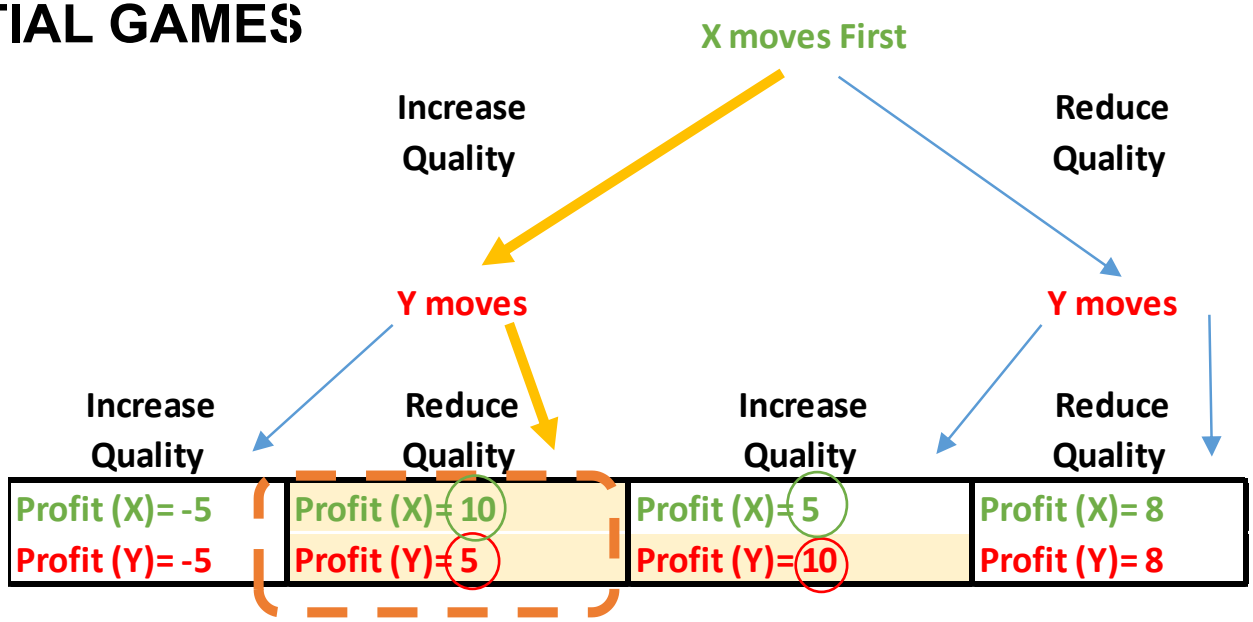
3. SEQUENTIAL GAMES

Below you can see a strategic decision of two firms – Firm X and Firm Y – illustrated in a decision tree. The two firms decide sequentially, Firm Y decides after Firm X made its decision.

(a) Please solve the game by backward induction and explain your reasoning.



3. SEQUENTIAL GAMES



We solve the game by looking at the second stage first. ***In the left (decision) knot****, Y can decide between losing -5 or earning 5 -> Y will choose to earn 5 and increase the quality.* ***In the right (decision) knot****, Y can decide between earning 10 or earning 8 -> Y will choose to earn 10 and will increase the quality.*

X knows how Y will behave *and therefore can decide between (a) increase the quality then Y will reduce quality and X will earn 10 and (b) reducing quality then Y will increase quality and X will earn 5.*

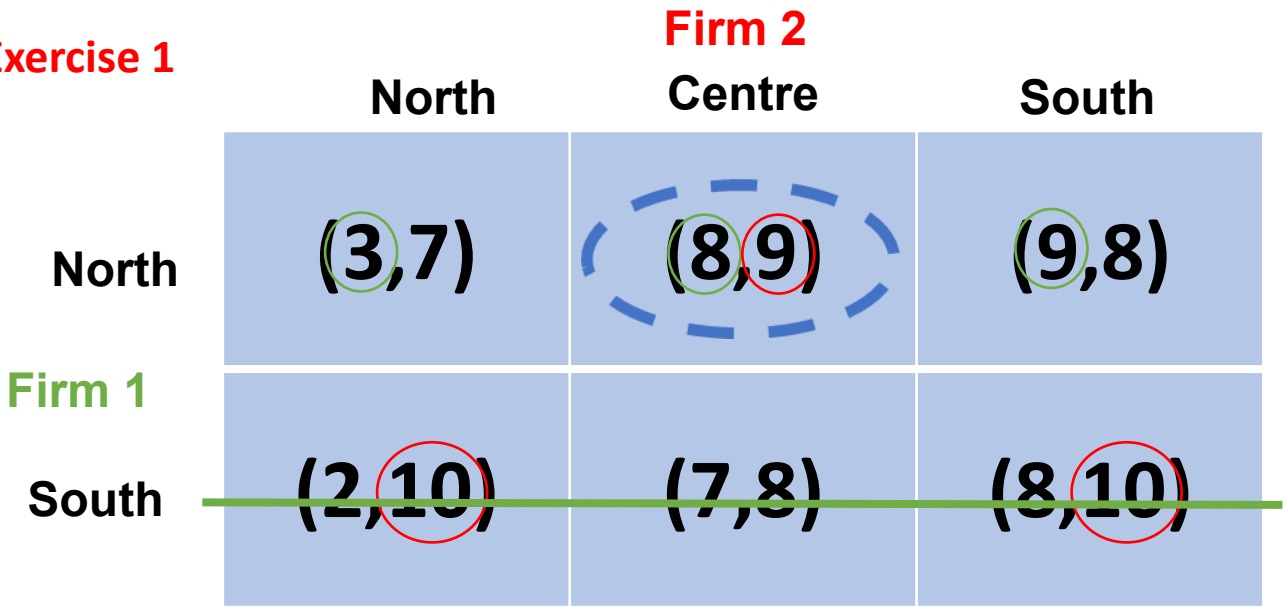
As X prefers to earn 10, X will reduce the price. Then Y decides and will not reduce the price. Therefore, in the end, X earns 10 and Y earns 5.

Exercise 1

Firm 1 \ Firm 2	North	Centre	South
North	(3,7)	(8,9)	(9,8)
South	(2,10)	(7,8)	(8,10)

1. Which is strictly dominated strategy and for which player?
2. In this game, which strategy (and for which player) can be eliminated as first one? Hence, which are the strategies for player 1 and Player 2 that will remain in the reduced form of the game?
3. Which is the Nash Equilibrium?
4. Are there any weakly dominated strategies in the game? Why are they called "weakly dominated"?

Exercise 1



3 > 2
8 > 7
9 > 8

9 > 7; 8
10 & 10 > 8

1. Which is strictly dominated strategy and for which player?

For FIRM 1 going NORTH is strictly dominated strategy because, no matter what firm 2 decides, they will always have a better outcome if they go north.

Exercise 1

		Firm 2		
		North	Centre	South
Firm 1	North	(3,7)	(8,9)	(9,8)
	South	(2,10)	(7,8)	(8,10)

2. In this game, which strategy (and for which player) can be eliminated as first one? Hence, which are the strategies for player 1 and Player 2 that will remain in the reduced form of the game?

For Firm 1 going SOUTH is a dominated strategy. Player 1 will always prefer going NORTH. So, going south can be eliminated as first one. Firm 2, assumes that Firm 1 will never chose going SOUTH.

		Firm 2		
		North	Centre	South
Firm 1	North	(3,7)	(8,9)	(9,8)

Exercise 1

Firm 1 \ Firm 2	North	Centre	South
North	(3,7)	(8,9)	(9,8)
South	(2,10)	(7,8)	(8,10)

3. Which is the Nash Equilibrium? (8,9)

Nash Equilibrium: For a two-player game, a Nash equilibrium is an outcome where player 2's strategy is the best response to player 1's strategy and player 1's strategy is a best response to player 2's strategy.

4. Are there any weakly dominated strategies in the game?

Why are they called "weakly dominated"?

Firm 1 has a weakly dominant strategy.

Strategy of Firm 1 is NEVER equally good or better than Firm 2.

7<3 10<2

9<8 8<7

9<8 10<8

Exercise 2

Firm 1 \ Firm 2	High MK	Low MK
High MK	(1,6)	(2,7)
Low MK	(4,1)	(3,2)

1. Which is the dominant strategy for Player 1?
2. Which is the dominant strategy Equilibrium?
3. Which is the Nash Equilibrium (NE)?
4. Is it Pareto efficient of NE? Why?
5. Which is the definition of NE?
6. Which is the difference compared with the dominant strategy Equibrium?

Example: Which market should a firm target? Dominant vs. Dominated Strategies



Firm 1 \ Firm 2	North	Centre	South
North	(1,0)	(1,2)	(0,1)
South	(0,3)	(0,1)	(2,0)

Additional Docs:

Game theory in action: The Golden Balls

https://www.youtube.com/watch?v=S0qjK3TWZE8



Watch the film→ Beautiful Mind

Film about John Nash

